

Energy-Guided Sampling for Controllable Text Generation

From Langevin Dynamics to Amortized Inference

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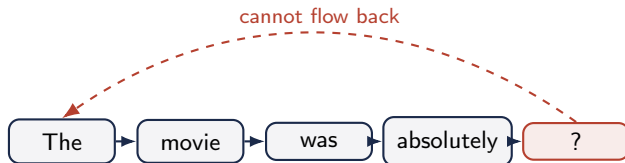
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M.Sc. Computational Linguistics

Thesis Defense

The problem autoregressive models cannot solve natively

- Autoregressive generation: one token at a time, left to right.
- A token is fixed the moment it is emitted.
No mechanism to revise an earlier token in light of a later one.
- Global properties (sentiment, factual constraint) live in the *whole* sequence.



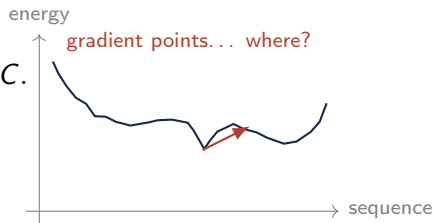
Control means steering the *completed* object. That is a global problem.

The energy-based promise, and its one assumption

Treat the frozen likelihood as an **energy** over sequences and sample from it:

$$p(\mathbf{x}) \propto \exp(-E(\mathbf{x})), \quad E = -\log p_{\text{LM}} + \lambda C.$$

- Add any constraint C at inference time.
- One frozen model, any objective.
Plug and play.



A jagged landscape: the local slope need not lead anywhere good.

The untested assumption:

the gradient of the frozen likelihood is a usable search direction on discrete text.

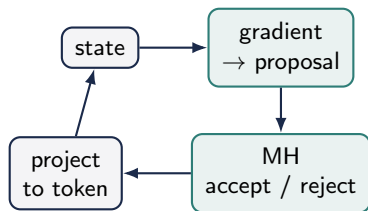
Research questions

- RQ1 Can the frozen likelihood be sampled with faithful Langevin dynamics on discrete text, and if not, why?
- RQ2 What does the Metropolis–Hastings correction do in each sampler?
- RQ3 Does amortizing with a GFlowNet escape the failure?
- RQ4 Can an additive constraint steer generation on this landscape?

The outcome is told up front: the thesis is an *explanation*, not a reveal.

Two samplers, implemented faithfully

- **DLS** (discrete): stays in token space; gradient only builds a proposal.
- **CLS** (continuous): relaxes into embedding space; projects back to tokens.
- **Metropolis–Hastings** correction: applied faithfully (usually omitted in the literature).
- Step size set by **oracle calibration** ($\epsilon : 10.5 \rightarrow 0.1$).



the step t
skips

One sampler loop, run to convergence, not stopped early.

5 energy functions, 145 configurations,
paired seeds, equivalence margin.

The central result: gradient vs. a norm-matched random direction

A three-way ablation isolates the gradient's *direction* from its *magnitude*.

Paired comparison, flagship config

mean difference +0.171 nats

95% CI $[-0.285, +0.619]$ (straddles zero)

Wilcoxon $p = 0.40$

detectable only if ≥ 0.652

Energy	policy	random
GPT-2 Large	6.474	6.430
GFN lb0-500	6.009	6.001
GFN lb0-2000	6.698	6.552
GFN lb1-500	6.008	5.338

Final KL (lower better). Differences small and inconsistent in sign.

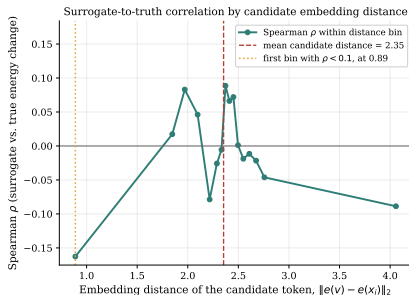
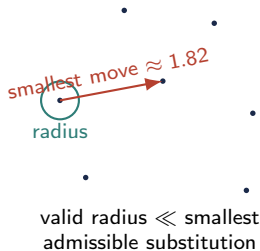
In the tested settings, the gradient gives no reliable advantage over noise of the same norm.

Why, part 1: the linearization fails before the first move

The proposal ranks tokens by a first-order surrogate. It is uncorrelated with the true energy change: $|\rho| < 0.06$ on all five energies ($n = 400,000$).

- Smallest token move: 1.82 (nearest-neighbour spacing).
- The surrogate is already uninformative at that distance.

The gradient is inapplicable, not merely imprecise.



Why, part 2: the correction breaks the continuous sampler

Decompose the Metropolis–Hastings acceptance for boundary-crossing moves:

The two terms

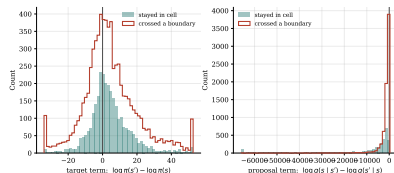
target log-ratio = +4.60 (moves are good)

proposal log-ratio = -1325 (kills them)

- The projected drift is not Lipschitz across a Voronoi boundary.
- CLS accepts within-cell moves $\approx 0\%$; DLS accepts them 100%.

Faithful correction disables CLS; it rescues DLS.

Metropolis–Hastings acceptance ratio, target and proposal terms, by whether the proposal crossed a Voronoi cell boundary



The energy is fine; its gradient is not

Score the *same* energy without its gradient (GPT-2 Large, masked recovery):

Method (no gradient of the energy)	mean KL
Ground-truth fill (floor)	0.00
Best Langevin configuration (uses the gradient)	~ 6.4
Gradient-free Gibbs sampler	6.69
Top- k rescore, one forward pass	4.43

Evaluate the energy: works. Differentiate it: does not.

The claim is narrowed to the gradient, not the energy.

GFlowNet amortization does not escape the failure

A GFlowNet only ever *evaluates* the reward, never differentiates it.

- Three failure modes: **capacity collapse**, **length collapse**, **reward hacking**.
- Tuning moved the energy by 31–57 nats. . .
- . . . yet Langevin sampling on the tuned energy is *indistinguishable* from the untuned base.

Changing the model through this objective did not make the gradient useful.

The failure is a property of the training objective, not one sampler.

The exact case: the final-token gradient is zero

Under causal attention with `inputs_embeds`, the gradient of the sequence log-likelihood w.r.t. the *final* token's input embedding is

$$\nabla_{\mathbf{e}_L} \log p_{\text{LM}}(\mathbf{x}) = 0 \quad (\text{exactly}),$$

while the energy *difference* between candidate final tokens is exactly the model's conditional log-ratio. Energy maximally usable; gradient provably useless.

Position	LM grad norm
final token	0.0000
second-to-last	> 0
middle	> 0

Langevin recovers 0% at all three positions; exact-energy methods 18–55%.

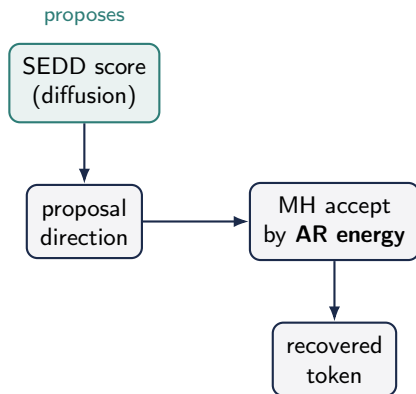
The positive control: a score-trained model supplies the direction

Same energy, same Metropolis chain, same sequences. Swap only the proposal direction: autoregressive gradient $\rightarrow$ diffusion (SEDD) score.

Exact recovery in the same exact-energy chain

autoregressive gradient: **0%**
left-conditional proposal: 23.5%
diffusion proposal (hybrid): **39%**

The missing direction was the training objective, not the sampler.



SEDD proposes; the frozen AR energy accepts.

0 $\rightarrow$ 39%.

Takeaways

- ❶ In the tested settings, the frozen autoregressive gradient is no better than a norm-matched random direction.
- ❷ The mechanism: linearization failure and, in continuous space, a Metropolis–Hastings breakdown at projection boundaries.
- ❸ The energy is usable: gradient-free rescoring and Gibbs recover coherent text; amortization inherits, not escapes, the problem; a score-trained model repairs it.

Evaluate the energy; do not differentiate it.

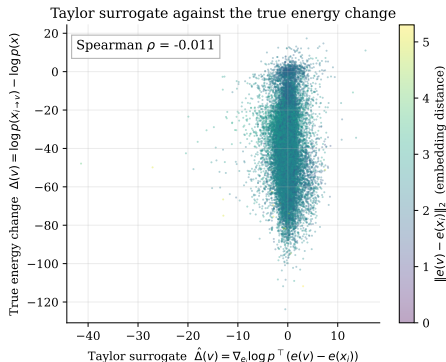
Training objectives determine landscapes.

Thank you.

Questions?

Backup: the linearization radius and self-term blindness

- The surrogate correlates with the *future* term ($\rho = 0.03$), never the *self* term ($\rho = -0.10$).
- Mean absolute self term 15.0 nats; future term 24.2 nats.
- The input-embedding gradient is structurally blind to the candidate's own left-context fit, which enters as a discrete index.



Backup: the Metropolis–Hastings chain

$$\alpha = \min \left\{ 1, \frac{p(\mathbf{s}') q(\mathbf{s} \mid \mathbf{s}')}{p(\mathbf{s}) q(\mathbf{s}' \mid \mathbf{s})} \right\}$$

- Target ratio: prefers good states. Proposal ratio: enforces reversibility.
- MALA needs a Lipschitz drift (Roberts and Tweedie 1996; Dalalyan 2017; Durmus and Moulines 2017).
- The projected target is piecewise constant; the differentiable pathway's drift jumps across a cell boundary, so the reverse density vanishes.

Backup: likelihood is a poor objective, yet used everywhere

- The **likelihood trap**: the lowest-energy text is the most degenerate (Holtzman 2020, reproduced here). Within-strategy repetition correlation up to +0.91.
- Resolution: likelihood ranks single-token substitutions well (used by the KL metric and Gibbs), but maximizing absolute sequence likelihood in free generation is degenerate.
- So using it as the KL metric, the reward, and the SEDD comparison is consistent: relative-to-reference fit, not absolute likelihood as quality.

Backup: MuCoLa / COLD, and why they appear to work

- The continuous sampler follows the COLD/MuCoLa mechanism faithfully, including the correction those works typically omit.
- Where they report success without the exact correction, it comes from an early-stopped biased optimizer plus post-hoc filtering, not from sampling the energy.
- The constrained *mucola* arm is that comparison, run at the energy level (final KL under the same base model), not the task level.

Backup: task design and corpus

- Masked-token recovery on WikiText-2 validation; the base GPT-2 Large is fine-tuned on ROCStories (out-of-domain test).
- In-domain ROCStories diagnostics reproduce the null and the likelihood trap: not a domain-shift artifact.
- The KL metric compares the recovered token's next-token conditional against the ground-truth-conditioned reference; synonyms score well.
- Llama-3 is a cross-architecture control, never on a shared numerical axis; there the gradient is reliably *worse* (CI [0.45, 2.09], $p = 0.015$).

Backup: why a GFlowNet, and the task-comparability caveat

- A GFlowNet samples in proportion to reward (not the mode), and never differentiates it, so a gradient pathology would be invisible to it.
- It reformulates infilling as left-to-right generation under a restructured prompt; the samplers do in-place bidirectional substitution.
- The comparison is therefore run at the energy level, not the task level; the left-to-right reformulation is not adopted for the samplers because it would dissolve the revision capability under study.